

Rolls-Royce Type R

Designed to win the Schneider Trophy seaplane race—and, at the same time, restore Rolls-Royce's position as Britain's top aero engine manufacturer—the R was based on the Buzzard V12 engine. The 2,239 cu in (36.7-liter) Buzzard produced 955 hp; running on special racing fuel, the R was ultimately developed to produce 2,783 hp.

Shaped cambox

The box is profiled to match the cowling and, through being exposed to the airflow, provides additional oil cooling.

External saddle studs
Additional studs, mounted outside the "V," have T-shaped feet, to spread the load when securing the cylinder blocks to the crankcase.

Coolant supply pipe to cylinder block

Propeller drive

Reduction gear housing
This gear set allows the engine to run at higher rpm, increasing power output.

A VERY SPECIAL ENGINE

Freed from the constraints applying to the development of military power units, the R was a highly specialized engine. Its external contours were shaped to match the slender lines of the Supermarine S.6 racer—the exposed cambox covers were profiled to match the cowling, and doubled as oil coolers. In almost trebling the standard production engine's power output, Rolls-Royce encountered numerous mechanical problems and ended up running its very noisy R test bed—which was serviced by three Kestrel V12 engines providing simulated airflow and cooling—for months on end.

Record breaker

Having powered the Supermarine S.6 seaplane to victory in the 1931 Schneider Trophy race, the R went on to help it set a new airspeed record of more than 400 mph (644 km/h). Only 19 R engines were built, and these were used in various record-breaking cars and boats.

Paired main bearing cross-bolts

Bolts run across the crankcase to increase the stiffness of the engine.

